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Developing Interactive Instructional Resources for STA258 Using Quarto

Statistics with Applied Probability

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STA258 slide website

Scan to explore the course resource

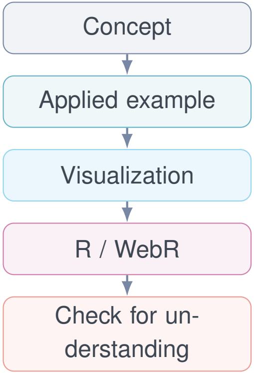
- Chapter landing pages
- Interactive Reveal.js slide decks
- WebR code blocks and exercises
- PDF views for static access

https://nishanmudalige.github.io/STA258_Slides/

Why this project?

A statistics course asks students to move repeatedly between:

- mathematical ideas,
- interpretation in context,
- visualization,
- computation in R, and
- practice and feedback.



Project aim: bring these steps into one coherent, browser-based learning experience.

Design goals

Clarity Keep statistical reasoning visible with concise explanations and deliberate visual hierarchy.

Interactivity Let students calculate, visualize, run R, and check understanding directly inside the slides.

Consistency Use a common structure, visual language, and interaction pattern across the course.

Applied context Anchor procedures in realistic examples before formal computation.

Reproducibility Keep source, computation, styling, and publishing within a maintainable workflow.

Accessibility Reduce software barriers by moving core interactions into the browser.

The design goal was not to make slides “more interactive” in isolation, but to make the learning sequence more coherent.

What we built

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Chapter-based course structure

Chapter landing pages and browser-based slide decks organized across the STA258 course sequence.

Statistical visualization

Reproducible R graphics, Plotly figures, and selected JavaScript widgets designed around applied examples.

Interactive computation

WebR, R demonstrations, solution reveals, and quiz interactions embedded directly within the learning materials.

A maintainable publishing system

Shared CSS, Git/GitHub workflow, rendered HTML/PDF access, and documentation for future maintainers.

The learning sequence

Definition or setup → Intuition → Applied example → Visuals / interactions → Check in R

- Repeating the same structure gives students a **predictable rhythm** from one topic to the next.
- Interactive elements appear where students can apply or check their understanding of the statistical idea.

Technical stack

- **Quarto** for authoring and rendering
- **Reveal.js** for browser-based presentations
- **R / knitr** for reproducible calculations and figures
- **WebR** for live R execution in the browser
- **Plotly** and JavaScript for interactive visualizations
- **HTML/CSS** for custom layouts and interaction design
- **Quiz plugins** for embedded formative checks
- **Git/GitHub** for collaborative version control
- **GitHub Pages** for deployment
- **PDF views** for printable / static access

The result is one workflow from:

source code → computation → interaction → published course resource

In-browser interactivity

Run R

WebR lets students execute selected R code without leaving the slide deck or installing software locally.

Reveal thinking

Fragment order controls when setup, calculation, and result appear during instruction.

Immediate checks

Embedded quizzes and solution reveals provide quick formative feedback.

Interaction principle: each click should move the statistical reasoning forward.

Example 1: Concept to visualization

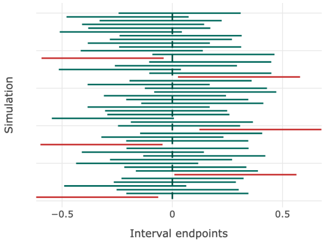
5.2 Interpreting Confidence

For a $C\%$ confidence interval:

In repeated sampling, about $C\%$ of intervals constructed using this method will capture the true parameter.

Do not say: “There is a 95% probability that this particular interval contains μ.”

The parameter is fixed. The random object is the interval produced by the sample.



Chapter 5 — interpreting confidence intervals, with a simulated-coverage Plotly figure

- The resource moves from **context** to **visual reasoning** before or alongside computation.

Example 2: Check in R

Example: Active Minutes

A smartwatch logs **daily active minutes** (time spent moving) for a random sample of 15 people. Active minutes are normally distributed with known $\sigma = 15$ minutes:

$79, 43, 58, 66, 101, 63, 79, 33, 58, 71, 60, 101, 74, 55, 88$

Construct a 95% CI for μ.

R Code   Start Over   ▷ Run Code

```

minutes <- c(79,43,58,66,101,63,79,3
xbar <- mean(minutes)
n <- length(minutes)
sigma <- 15
zstar <- qnorm(0.975)
ci <- xbar + c(-1, 1) * zstar * sigm

xbar
ci
```

$\bar{x} = 68.6$, so the interval is approximately $(61.01, 76.19)$. We are 95% confident that the population mean number of daily active minutes is between about 61 and 76.

Chapter 5 — a worked example with a live WebR editor beside it

- Connects the hand-worked statistical method directly to its implementation in R.
- Especially valuable because STA258 tutorials require R, and this support carries into upper-year statistics courses.

Example 3: Formative feedback

Question

A sample has $\bar{x} = 42$, $n = 36$, and known $\sigma = 12$. For a 95% CI, what is the margin of error?

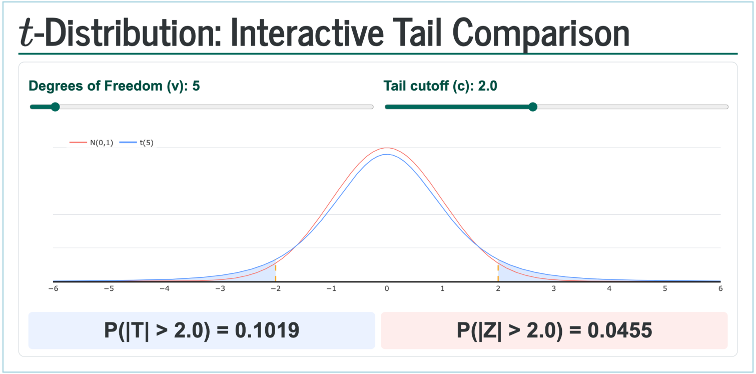
- 4.00
- 1.96
- 7.84
- 3.92

Check Reset Prev Next

Chapter 5 — an embedded quiz with immediate checking and explanations

- The goal is not simply to reveal an answer, but to make the reasoning and common misconception visible.

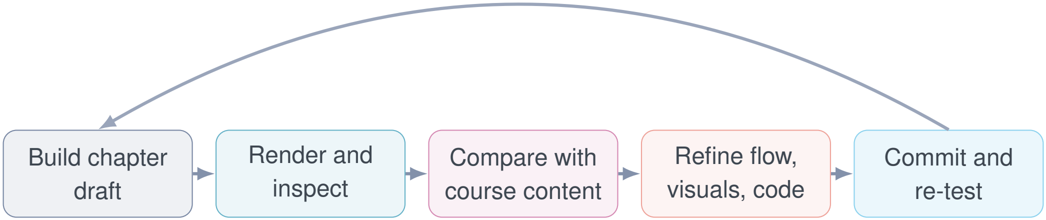
Example 4: Interactive statistical visualization



Chapter 4 — a t-distribution tail explorer with live sliders for ν and the cutoff

- The interaction reinforces what changes, what stays fixed, and what the graphical objects represent statistically.

An iterative summer development process



- Refinement focused on **genuine instructional discrepancies**: incorrect formulas, broken interaction order, misleading wording, inconsistent examples, and serious layout issues.
- The workflow also standardized chapters without changing content simply for cosmetic uniformity.

What consistency work looked like

Instructional consistency

- Example flow and fragment order
- Notation and final-answer precision
- R checks that match the worked example
- Conclusions stated in context
- Chapter-to-chapter navigation

Visual / technical consistency

- Fixed scaling and overflow
- Consistent Plotly presentation
- Readable axes and labels
- Solution / reset behavior
- Shared CSS and reusable layouts

Consistency became a **course-level design problem**, not just a collection of isolated slide edits.

Designed for maintainability

- Shared styling is centralized rather than duplicated across slides.
- Chapter files follow a predictable structure.
- A help manual documents rendering, navigation, WebR, quiz behavior, Git workflow, and common errors.
- The repository structure separates editable source from rendered output.

`chapterN-slides.qmd`

shared `styles.css`

`_extensions/` + WebR

HELP / maintenance guide

rendered `docs/`

What the project enables

For students

A single browser-based path connecting explanation, visualization, computation, and formative practice.

For instructors

Reusable teaching sequences with controlled reveal order, consistent examples, and integrated R demonstrations.

For future developers

A documented structure that can be updated chapter-by-chapter without rebuilding the entire system.

The contribution is both **instructional** and **infrastructural**:
the content and the system used to deliver it are developed together.

Limitations

- The resource is still in development and has not yet been fully evaluated through live course deployment.
- Browser-based interactivity introduces dependencies on rendering, JavaScript, and WebR behavior across devices.
- Interactive slides require more maintenance than static lecture material because code, styling, and pedagogy must remain synchronized.
- Accessibility and mobile responsiveness require continued testing as the resource evolves.

Future work

Fall 2026
Testing, refinement, and final consistency review

→

Ongoing
Accessibility, browser testing, maintenance, and content updates

→

Winter 2027
Live deployment and classroom feedback

- Use classroom feedback to determine which interactions genuinely support learning and which should be simplified.
- Continue developing the resource as a maintainable course platform rather than a one-time slide deck.

Takeaways

1. **Quarto provides a single authoring environment** for text, mathematics, R, HTML/CSS, and interactive presentation components.
2. **Interactivity is most useful when it follows the statistical reasoning**, rather than being added as a separate feature.
3. **Consistency across chapters matters pedagogically**: students can focus on the concept when the structure and interaction patterns are predictable.
4. **Maintainability is part of instructional design**: future course updates should be possible without reconstructing the system.

Acknowledgements

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Thank you!

Questions or comments?

`https://nishanmudalige.github.io/STA258_Slides/`

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